FOR EDITH · OBSTETRICS & GYNAECOLOGY JUNIOR CLERKSHIP

Vaginal Delivery: Equipment, Sutures & Simulated Delivery

Clinical skill | Week 3 · Medical disorders & PPH | Dr Kavuma, Sr Kiggundu

Know the delivery pack and suture materials by name and purpose, conduct a clean, controlled spontaneous vaginal delivery with AMTSL and newborn care, perform an episiotomy only when indicated, and classify and repair perineal tears correctly, recognising and referring obstetric anal sphincter injuries (third- and fourth-degree tears).

Must know before the ward round

1. Standard delivery pack - 2 artery forceps, cord scissors, episiotomy scissors, needle holder, toothed dissecting forceps, sponge-holding forceps, gallipot, kidney dish, cord ties/clamps, gauze and drapes.
2. Before delivery - confirm second stage, FHR, maternal pulse/BP, risk factors (big baby, previous CS, twins, meconium), oxytocin 10 IU drawn up and bag-and-mask tested.
3. Controlled delivery of the head (flex, then let it extend slowly between contractions while she pants) is the best way to prevent perineal tears.
4. Palpate for a second twin, then oxytocin 10 IU IM within 1 minute; delay cord clamping 1-3 minutes in a vigorous baby.
5. Fourth-stage sweep - uterine tone, measured blood loss, placenta and membranes (3 cord vessels), perineum/vagina/cervix, rectal exam, vital signs, bladder, baby; tone and bleeding every 15 min for 2 h.
6. Firm uterus + continuing bleeding = trauma or retained tissue - look in good light; a vessel running to a torn membrane edge means a retained succenturiate lobe.
7. Episiotomy is NOT routine (WHO). If needed - mediolateral, about 60 degrees from the midline at crowning, after infiltrating 1% lignocaine (aspirate first; max about 3 mg/kg without adrenaline).
8. Tear classification (RCOG) - 1st skin/vaginal epithelium; 2nd perineal muscles; 3a less than 50% of external anal sphincter (EAS); 3b more than 50% EAS; 3c EAS and internal anal sphincter; 4th sphincter complex plus anorectal mucosa.
9. Always do a rectal examination before and after any perineal repair - to detect sphincter injury and to check no suture went through the rectal mucosa.
10. Second-degree repair - rapidly absorbable polyglactin (e.g. Vicryl Rapide 2-0), continuous non-locking for vaginal mucosa and muscles, subcuticular skin; first stitch 1 cm above the apex.
11. OASI - repair in theatre with good light and anaesthesia by an experienced doctor; PDS 3-0 (or Vicryl 2-0/3-0) for sphincters; broad-spectrum antibiotics and laxatives; follow-up for incontinence.
12. Count swabs, needles and instruments before and after; a retained swab is a preventable cause of sepsis and fistula.

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1. Purpose and Indications

This is a **skills topic**. By the end of the rotation you should be able to: name every instrument in a delivery pack and say what it is for; choose the right suture for the right tissue; conduct a clean, controlled normal delivery on a mannequin and (supervised) on the ward; decide when an episiotomy is truly needed and cut and repair one; and **classify perineal tears correctly**, repairing first- and second-degree tears and **recognising third- and fourth-degree tears** so they go to the right person in theatre.

Why it matters in Uganda: perineal trauma is common, especially in primigravidae and with instrumental deliveries. **Missed anal sphincter injury** leaves women with lifelong faecal incontinence, a hidden disability that causes shame and marital breakdown. Poor asepsis and retained swabs lead to **puerperal sepsis**. Most of this is prevented by **good technique and a careful examination after birth**.

> CLINICAL PEARL
> The one thing never to forget: after every vaginal birth, **inspect the perineum, vagina and cervix in good light and do a rectal examination** before you decide what the tear is. You cannot classify what you have not looked at.

Burden and Uganda context

- Most vaginal births involve some trauma; about 60-70% of women need suturing in many series. OASI occurs in roughly 1-6% of vaginal births depending on how carefully it is looked for; it is under-diagnosed where examination is poor.
- **Risk factors for OASI**: primiparity, birth weight over 4 kg, OP position, **instrumental delivery (especially forceps)**, prolonged second stage, **midline episiotomy**, shoulder dystocia, Asian ethnicity, induction, precipitate delivery.
- **Ugandan realities**: high patient volumes with one midwife per shift, **chromic catgut** still found in some units (more pain and tissue reaction than synthetic sutures), inadequate light, and variable availability of local anaesthetic. Women living with **HIV** and those with **FGM** scarring (in some communities of the north-east) need special care.
- **Obstetric fistula** (VVF/RVF) is a different injury (pressure necrosis from obstructed labour), but a badly repaired fourth-degree tear or a suture through the rectum can also cause a **rectovaginal fistula**.

Indications

- **Conducting a delivery**: every spontaneous vaginal birth with a longitudinal lie, cephalic presentation and no indication for operative delivery (see the pre-delivery checks in section 4).
- **Repair**: first-degree tears that bleed, gape or distort anatomy; all second-degree tears and episiotomies.
- **Episiotomy (restrictive use only)**:
  - **Fetal distress** in the second stage, when birth needs to be hastened and the perineum is the obstruction
  - **Assisted vaginal delivery** (forceps always; vacuum only if the perineum is tight)
  - **Rigid or scarred perineum** causing delay (e.g. previous third-degree tear repair; FGM type III requires anterior defibulation instead)
  - **Shoulder dystocia** to allow internal manoeuvres (it does not release the impacted shoulder itself, but gives room for your hand)
  - **Breech delivery** in selected cases

2. Contraindications and Precautions

Episiotomy is not indicated: routinely in primigravidae; to prevent tears (it does not); for preterm babies routinely.

When a student should NOT repair, and must call a senior

- Any **third- or fourth-degree tear** or uncertainty about the sphincter
- **Heavy bleeding** from a tear that you cannot see the apex of, or cervical/high vaginal tears
- A **haematoma**
- Poor light, no assistant, or the woman is unstable (manage PPH first)

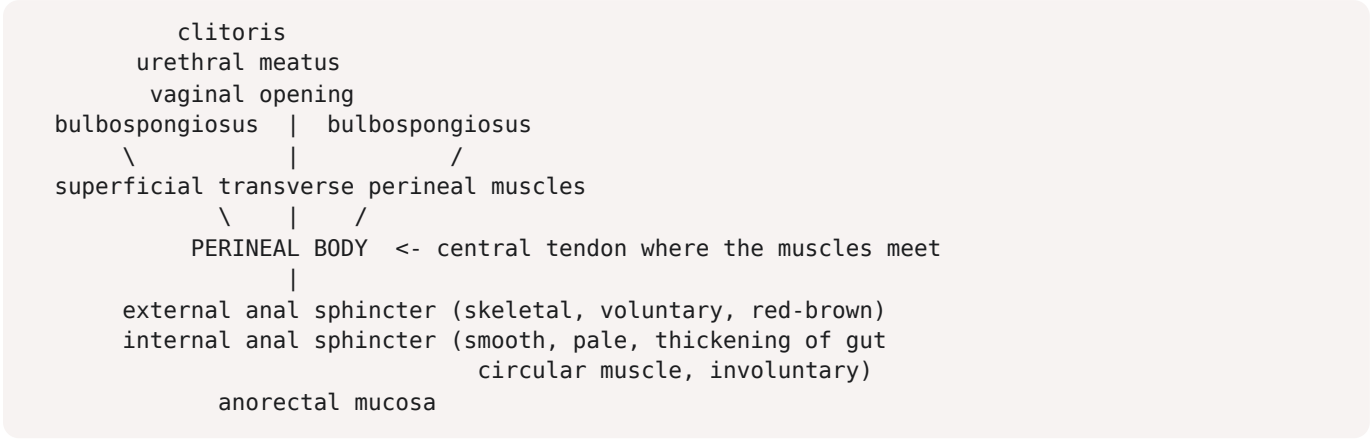
When a normal delivery by you is not appropriate (call a senior): breech, transverse, face or brow found late; head still 2/5 or more at full dilatation; OP/OT with a slow second stage; moulding +++; abnormal FHR; exhausted, bleeding or BP 160/110 or more; rigid, scarred or FGM type III perineum.

Other precautions

- **No ergometrine** in hypertension, pre-eclampsia or cardiac disease.
- **Do not give oxytocin** until a second twin has been excluded.
- **HIV**: avoid unnecessary episiotomy; safe sharps practice (blunt-tip needles if available, never pass sharps hand-to-hand).
- **Lignocaine**: aspirate before every injection; keep within the maximum dose.
- **Never fundal pressure** and never traction on the neck.
- **Cord traction only during a contraction with counter-traction** (uterine inversion risk).

3. Relevant Anatomy and Normal Values

The perineum: why the repair is done the way it is



- The **perineal body** is the keystone of the pelvic floor. The perineal muscles (bulbospongiosus and superficial transverse perineal) insert here. A good second-degree repair re-attaches these muscles to rebuild the perineal body; a poor repair leaves a gaping introitus, dyspareunia and prolapse later.
- The **external anal sphincter (EAS)** is striated muscle, dark red, about 3-4 cm long; when torn, its ends retract into their capsule (so you must grasp them with Allis forceps). It provides **voluntary** continence (squeeze).
- The **internal anal sphincter (IAS)** is pale, smooth muscle, a continuation of the rectal circular muscle; it provides **resting tone** and most passive continence (about 70-80%). Missed IAS injury causes passive soiling and flatus incontinence.

- The **pudendal nerve (S2-S4)** supplies perineal sensation and the EAS; a **pudendal block** at the ischial spines anaesthetises the perineum.
- **Blood supply**: the internal pudendal artery. The perineum heals well because it is vascular, but it bleeds readily.

Episiotomy types

Type	Description	Comments
Mediolateral	From the posterior fourchette directed laterally, **about 60 degrees from the midline at crowning** (ends up about 45 degrees after delivery)	**Standard in Uganda and the UK**. Lower risk of extension into the sphincter
Midline (median)	Straight down towards the anus	Easier to repair, less pain, but **higher risk of third/fourth-degree extension**; not used in our setting
Lateral / J-shaped	Starts lateral or curves	Not recommended

Normal values

Item	Normal
FHR in second stage	**110-160 bpm**, checked **every 5 min** (or after each contraction)
Blood loss	**Under 500 mL** (measured)
Shock index (pulse ÷ SBP)	**Under 0.9**
Uterus after delivery	Firm, contracted, at or below the umbilicus
Placenta	Complete cotyledons and membranes; cord with **3 vessels** (2 arteries, 1 vein); central or eccentric insertion
Voiding after delivery	Within **6 h**
Retained placenta	Not delivered **30 min** after birth
Cord ties	About **2 cm and 5 cm** from the umbilicus

4. Equipment and Preparation

The delivery pack (typical Ugandan labour ward)

Item	Purpose
2 artery forceps (Kocher's or Spencer Wells)	Clamp the cord (and bleeding vessels)
Cord scissors (blunt-ended, curved)	Cut the cord between clamps
Episiotomy scissors (Braun-Stadler; angled, blunt tip)	Cut an episiotomy (the blunt tip protects the baby's head)
Needle holder (Mayo-Hegar)	Hold the needle during repair
Toothed dissecting forceps	Hold tissue during repair
Sponge-holding forceps (Rampley's/ring forceps)	Clean the vulva, remove membranes, inspect the cervix
Gallipot	Antiseptic solution (e.g. chlorhexidine or povidone iodine)
Kidney dish	Receive the placenta, used instruments

Item	Purpose
Cord ties or plastic cord clamps	Tie the cord at about 2 cm and 5 cm from the umbilicus
Gauze swabs, cotton wool, sanitary pads	Cleaning, perineal support, counting
Sterile drapes/towels (at least 2 for the baby)	Drying and wrapping the baby; creating a sterile field
Sterile gloves (and elbow-length gloves for manual removal)	Asepsis
Urinary catheter (Nelaton or Foley)	Empty the bladder in the second or third stage if needed

Separate items to have ready: oxytocin 10 IU (fridge), 2 mL and 10 mL syringes and needles, 1% lignocaine, suture materials, a **speculum (Sims or Cusco)** and a good light for inspection, a **penguin suction bulb** and a **bag and mask (sizes 0 and 1)**, a clock with a second hand, a baby cap, a weighing scale, vitamin K, tetracycline eye ointment, chlorhexidine gel, a **calibrated under-buttock drape** where available, and a sharps container.

Suture materials: principles

Property	Options	Clinical meaning
Absorbable vs non-absorbable	Absorbable: polyglactin 910 (Vicryl, Vicryl Rapide), polyglycolic acid (Dexon), polydioxanone (PDS), chromic catgut. Non-absorbable: nylon, silk, polypropylene (Prolene)	**Perineal repairs use absorbable sutures** so they do not need removal
Natural vs synthetic	Natural: catgut (sheep/cattle intestine), silk. Synthetic: Vicryl, PDS, nylon	Synthetics cause **less tissue reaction and less pain**; catgut is absorbed by inflammation (proteolysis)
Braided (multifilament) vs monofilament	Braided: Vicryl, silk. Monofilament: PDS, nylon, Prolene	Braided: easier to tie, holds knots, but can harbour bacteria. Monofilament: slides through tissue, less infection
Absorption time	Vicryl Rapide: loses strength in about 10-14 days, absorbed in about 6 weeks. Vicryl: absorbed in about 8-10 weeks. PDS: strength for about 6 weeks, absorbed in about 6 months	Rapid-absorbing sutures are ideal for the perineum (less pain, fewer suture removals). **PDS** for the sphincter because it needs support for longer
Size (gauge)	Bigger number followed by "-0" = thinner (3-0 thinner than 2-0 thinner than 0 thinner than 1)	2-0 for vaginal mucosa and muscle; 3-0 for skin and sphincter
Needle	**Round-bodied (taper)**: pushes tissue apart (muscle, mucosa, bowel). **Cutting / reverse cutting**: cuts through tough tissue (skin)	Use a **round-bodied** needle in the perineum; a large curved needle (e.g. 35 mm half-circle) is easier for the vagina

> **UGANDA CONTEXT**
> Many Ugandan units still stock **chromic catgut 2-0 or 0**. It works but causes more pain and needs removal more often. **Polyglactin (Vicryl or Vicryl Rapide) 2-0** is preferred where available (WHO, RCOG, Cochrane evidence). Use what your ward has, but know why synthetic is better.

First 5 minutes: when you are called "she is pushing"

1. **Confirm the second stage**: urge to push, anal pouting, head visible, or full dilatation on VE.
2. **FHR now** (for 1 minute after a contraction): 110–160 bpm. Abnormal → call a senior and prepare for assisted delivery and resuscitation.
3. **Maternal pulse and BP**: BP 160/110 or more → no ergometrine, treat severe hypertension; pulse 120 or more → look for bleeding, sepsis or exhaustion.

4. **Quick risk scan of the chart**: previous CS, big baby, prolonged second stage, meconium, twins, anaemia, HIV, pre-eclampsia. Any one of these → a senior should be present or near.
5. **Set up**: gloves and PPE, delivery pack open, **oxytocin 10 IU drawn up**, newborn corner ready with **bag and mask tested**, clock visible, someone to call for help.

Focused history: what to ask, and why

Before delivery (a 1-minute checklist; each answer changes what you prepare):

Question	Why it matters
Parity, previous deliveries and perineal trauma (third-degree tear, repair, incontinence, fistula repair)	Previous OASI or fistula repair: a senior should decide the mode of delivery and be present; watch for recurrence
Previous CS	Watch for scar rupture; delivery where theatre is available
Estimated fetal weight, diabetes, prolonged second stage	Macrosomia: prepare for **shoulder dystocia** (HELPERR, extra help, a senior in the room); OASI risk
Twins, or uncertain number	Do not give oxytocin until a second baby is excluded
Meconium, abnormal FHR, preterm	Skilled resuscitator at the birth
HIV status, viral load	Avoid unnecessary episiotomy; safe sharps handling; infant prophylaxis ready
Hypertension/pre-eclampsia, cardiac disease	No ergometrine
Anaemia, previous PPH, grand multiparity	IV line in place, blood grouped, active PPH prevention
FGM	May need anterior defibulation; scarred perineum tears unpredictably
Allergies (lignocaine, iodine, latex)	Choose safe agents
Bleeding disorders, anticoagulants	Haematoma risk

After delivery: ask about **pain** (severe perineal or rectal pain = haematoma), **dizziness** (blood loss), **urge to pass urine** (retention), and feeling of **anal urgency or loss of control** (missed OASI).

Pre-delivery examination (Hutchison order)

1. **Introduce yourself, consent, privacy, companion**; **hand hygiene**.
2. **General inspection**: distress, exhaustion, pallor, bleeding; **vital signs** (pulse, BP).
3. **Obstetric abdominal examination**: inspection, fundal height, **lie**, **presentation**, **descent in fifths**, contractions, **auscultation of the FHR** for 1 minute after a contraction.
4. **VE** (if not recently confirmed): full dilatation, position, moulding, caput, station.
5. **Perineum**: stretching normally, or rigid/scarred/FGM.

Check	Normal to proceed	Stop and call a senior if…
Lie and presentation	Longitudinal, cephalic	Breech, transverse, face/brow found late
Descent	0/5–1/5 palpable, head visible or low	Head still 2/5 or more at full dilatation
Position (VE)	OA (posterior fontanelle anterior)	OP/OT with slow second stage
Moulding / caput	0 to +	+++ (CPD)
FHR	110–160, no late decelerations	Abnormal → expedite (vacuum if criteria met)
Maternal state	Pulse, BP normal; able to push	Exhausted, bleeding, BP 160/110 or more
Perineum	Stretching normally	Rigid, scarred or FGM type III (needs defibulation)

Consent and positioning

- **Consent** for delivery, any episiotomy (explain why it is needed) and any repair.
- **Position** the woman as she prefers (semi-recumbent, left lateral, squatting, all fours). For the mannequin OSCE, semi-recumbent with legs apart. For **repair**: lithotomy (or at the edge of the bed) with **good light**.

5. Procedure Step by Step

A. Preparing for delivery

1. **Greet and consent**: introduce yourself, explain, ask permission; ensure privacy and a companion.
2. **Hand hygiene**, apron, eye protection; **sterile gloves**.
3. **Check equipment**: open the delivery pack on a clean trolley; oxytocin 10 IU drawn up; **newborn corner** ready (warm, towels, cap, bag and mask checked, suction bulb, clock).
4. **Position** the woman as she prefers.
5. **Clean the vulva and perineum** with antiseptic (cotton swabs held in sponge forceps: from above downwards, front to back, one stroke per swab, inner to outer). Drape under the buttocks.
6. **Empty the bladder** if full and she cannot void.

B. Conducting the delivery (simulated or real)

1. **Confirm full dilatation** (if not already confirmed) and monitor FHR every 5 min.
2. **Encourage spontaneous pushing** with contractions; rest between.
3. As the head distends the perineum, apply a **warm compress** and support the perineum with a pad in the right hand (the "hands-on" technique).
4. **Keep the head flexed**: place the fingers of the left hand gently on the occiput to control the speed of delivery, not to hold it back forcefully.
5. **At crowning**: ask her to **stop pushing and pant** (or blow), so the head delivers slowly between contractions.
6. If an episiotomy is indicated, perform it now (see C).
7. **Deliver the head** by extension: occiput, then forehead, face and chin sweep over the perineum.
8. **Wipe the baby's face** (mouth and nose) with gauze; do **not** routinely suction.
9. **Check for the cord around the neck**: loose, slip it over the head (or let the body somersault through); tight, clamp twice and cut between (only if necessary).
10. **Wait for restitution and external rotation** with the next contraction.
11. **Deliver the anterior shoulder** with gentle downward traction (hands over the parietal bones and cheeks, **never pulling on the neck**), then the **posterior shoulder** with gentle upward traction, while watching the perineum.
12. **Deliver the body** by lateral flexion (following the curve of the birth canal) and place the baby on the mother's abdomen. **Note the time of birth.**
13. **Dry thoroughly**, assess breathing, keep skin-to-skin, cover. If not breathing: Golden Minute (clear airway if needed, stimulate, then ventilate; see the neonatal resuscitation topic).
14. **Palpate the abdomen to exclude another baby**, then give **oxytocin 10 IU IM** within 1 minute of birth. Watch the blood loss from the moment the baby is born (use a **calibrated drape** where available).
15. **Delay cord clamping 1-3 minutes** (if the baby is breathing), then clamp/tie at about 2 cm and 5 cm and cut between with cord scissors, shielding the cut with gauze.
16. **Controlled cord traction** with counter-traction when the uterus contracts; deliver the placenta and membranes.
17. **Massage/check the uterus**; check tone every 15 min for 2 hours.
18. **Examine the placenta** (cotyledons, membranes, vessels, cord insertion, 3 vessels).

19. **Inspect the perineum, vagina and cervix**; perform a rectal examination; repair tears.
20. **Estimate blood loss**, clean the woman, sanitary pad, make her comfortable, help her breastfeed.
21. **Dispose** of sharps, decontaminate instruments (0.5% chlorine), hand hygiene, **document**.

C. Performing a mediolateral episiotomy

1. **Explain** why it is needed and get consent.
2. **Infiltrate 1% lignocaine** (about **10 mL**) along the planned line: insert the needle at the fourchette, direct it about **60 degrees from the midline** (towards the ischial tuberosity on the mother's right usually), with two fingers of the other hand in the vagina between the fetal head and the perineum to protect the head. **Aspirate before injecting** (to avoid intravascular injection), inject while withdrawing. Fan into the vaginal mucosa, skin and muscle.
3. Wait **about 2 minutes** for the anaesthetic to work (test with forceps).
4. At **crowning** (when the perineum is thinned and 3-4 cm of head is visible during a contraction), insert two fingers between the head and perineum and cut with **episiotomy scissors in a single, decisive cut of about 3-4 cm**, starting at the midpoint of the fourchette and directed **60 degrees** away from the midline.
5. **Control the head** carefully after the cut (it can deliver suddenly and extend the incision).
6. If there is bleeding before birth, apply pressure with a swab.

> **DRUG DOSING**
> **Lignocaine (lidocaine) 1%** = 10 mg/mL. Maximum dose without adrenaline is about **3 mg/kg** (about 18-20 mL for a 60 kg woman). 10 mL is usually enough for an episiotomy or second-degree repair. Always **aspirate before injecting**. Signs of toxicity: perioral tingling, metallic taste, tinnitus, drowsiness, seizures, arrhythmias; stop injecting and call for help.

D. The fourth-stage sweep (examination after delivery)

In this order, within the first 15–30 minutes, then tone and bleeding every 15 minutes for 2 hours:

1. **Uterine tone**: firm, at or below the umbilicus?
2. **Blood loss**: measured (calibrated drape, weighed pads).
3. **Placenta and membranes**: lay out on a flat surface; maternal side, fetal side, membranes, cord vessels.
4. **Inspection in good light**: labia, periurethral area, vaginal walls (sulci), perineum, and the **cervix** with a speculum if bleeding persists with a contracted uterus. **Assess the apex** of vaginal tears (if you cannot see it, you cannot repair it safely).
5. **Rectal examination** (every time): finger in the rectum, thumb on the perineum, "**pill-rolling**" to feel the sphincter thickness; look for a **rectal mucosal tear**.
6. **Vital signs**: pulse, BP, shock index (bleeding may be from atony, not the tear).
7. **Bladder**: palpable? voided?
8. **Baby**: breathing, warm, skin-to-skin, breastfeeding.

E. Repairing an episiotomy or a second-degree tear

Principles: **good light, adequate analgesia, an assistant if possible, identify the apex, restore anatomy layer by layer, avoid dead space, do not over-tighten** (tissues swell).

1. **Explain and get consent.** Position in lithotomy (or at the edge of the bed) with good light.
2. **Clean** the area; drape; **check the placenta is complete** and the uterus contracted before starting.
3. **Inspect systematically** (vagina, cervix if bleeding, perineum) and **do a rectal examination** to assess the sphincter (pill-rolling the EAS between the finger in the rectum and the thumb on the perineum).
4. **Re-infiltrate lignocaine** if needed (check the total dose).

5. Insert a **tampon/gauze roll with a tail** high in the vagina if blood from above obscures the view (and **remember to remove it**; count it).
6. **Vaginal mucosa**: identify the **apex**; place the first stitch **about 1 cm above the apex** (to catch retracted bleeding vessels). Close the vaginal mucosa with a **continuous non-locking** suture of **2-0 polyglactin** down to the hymenal ring, bringing the edges of the fourchette together.
7. **Perineal muscles**: bring the needle through the fourchette into the perineal wound and close the **muscle layer** (bulbospongiosus and transverse perineal) with the same continuous non-locking suture (or interrupted sutures), ensuring the perineal body is rebuilt and there is **no dead space**. Deep muscle may need a second layer.
8. **Skin**: close with a **continuous subcuticular** suture (just under the skin) ending with a buried knot in the vagina behind the hymen; or interrupted 2-0/3-0 sutures if skill is limited.
9. **Rectal examination again** to make sure **no suture has gone through the rectal mucosa** (if so, remove it and redo; otherwise a rectovaginal fistula can form).
10. **Remove the tampon/gauze roll**; **count swabs, needles and instruments**.
11. Clean, apply a pad, make her comfortable; document the degree of tear, repair, suture used, EBL and counts.

> **NOTE**
> The **continuous non-locking technique** for all layers causes **less pain** at 10 days than interrupted sutures and uses less suture material (Cochrane). Locking ("blanket") stitches strangulate tissue and increase pain.

F. Repairing a first-degree tear

- Small, non-bleeding, well-aligned first-degree tears and labial grazes **can be left unsutured** (after discussion with the woman), as they heal well.
- Suture if bleeding, if the edges gape, or if anatomy is distorted, using 2-0 or 3-0 polyglactin (continuous for skin/mucosa).

G. Third- and fourth-degree tears (OASI): what the student must know

1. **Recognise** it: gaping perineum with the dark red, retracted ends of the EAS; **pale IAS**; torn anorectal mucosa. The finger in the rectum shows a thin or absent sphincter anteriorly.
2. **Call the senior doctor/obstetrician.** Do not attempt the repair.
3. Apply pressure with a pad to control bleeding; keep her comfortable; document.
4. **Repair in theatre** under **spinal or general anaesthesia** (relaxes the sphincter so the ends can be approximated), good light, proper instruments (Allis forceps).
5. **Technique** (by the experienced surgeon): anorectal mucosa with **3-0 polyglactin** (continuous or interrupted), **IAS separately with 3-0 PDS** (or 2-0 polyglactin) end-to-end, **EAS** with **3-0 PDS** or 2-0 polyglactin by **end-to-end (3a, some 3b) or overlapping (full thickness) technique**, then the perineal muscles, vagina and skin as for a second-degree repair.
6. **Post-repair care** (RCOG):
    - **Broad-spectrum antibiotics** (for example intraoperative cefuroxime 1.5 g IV + metronidazole 500 mg IV, then an oral course such as amoxicillin-clavulanate + metronidazole for about 5-7 days, per local protocol), to reduce infection and wound breakdown.
    - **Stool softeners/laxatives** (lactulose 10-15 mL twice daily for about 10 days) so that the first stools do not disrupt the repair; avoid constipating codeine.
    - **Analgesia** (paracetamol, NSAID).
    - **Urinary catheter** for about 12-24 h (pain and swelling cause retention).
    - **Physiotherapy** (pelvic floor exercises) and **follow-up at 6-12 weeks** for continence; counsel about future deliveries.

UGANDA CONTEXT

At an HC III or HC IV without an experienced surgeon, a woman with a suspected OASI should be **referred** (after haemostasis with a pressure pack) to a hospital where a skilled obstetrician can repair it in theatre. A good primary repair offers the best chance of continence; later secondary repairs are much less successful.

6. Findings and Interpretation

Perineal tear classification (RCOG Green-top Guideline 29, used internationally)

Degree	Structures injured
First	Injury to **perineal skin and/or vaginal mucosa** only
Second	Injury to the **perineal muscles** (bulbospongiosus, transverse perinei) but **not the anal sphincter**
Third	Injury to the **anal sphincter complex**
3a	**Less than 50% of the external anal sphincter (EAS)** thickness torn
3b	**More than 50% of EAS** thickness torn
3c	**Both EAS and internal anal sphincter (IAS)** torn
Fourth	Injury to the **anal sphincter complex (EAS and IAS) and the anorectal mucosa**
Rectal buttonhole	Tear of the rectal mucosa with an **intact sphincter**; not a fourth-degree tear, but must be repaired and followed up

EXAM FAVOURITE

If you are unsure whether a tear is 3a or 3b, **classify it as the higher degree** (RCOG). Under-classification leads to inadequate repair.

OASI = obstetric anal sphincter injury (third- and fourth-degree tears).

Other lower genital tract injuries

- **Labial and periurethral tears**: often bleed; repair if bleeding or anatomy distorted (catheterise if near the urethra).
- **Vaginal wall (sulcus) tears**: high up in the lateral fornices, often after instrumental delivery; can bleed heavily.
- **Cervical tears**: bleeding with a well-contracted uterus; inspect with a speculum and sponge forceps.
- **Vulval/paravaginal haematoma**: severe perineal pain and a tense bluish swelling after delivery; may need evacuation.

Fourth-stage sweep: what each finding means

Order	Check	Normal finding to document	Abnormal finding and action
1	**Uterine tone**	Firm, contracted, at or below the umbilicus	Soft, boggy, above the umbilicus → massage, repeat uterotonic, empty bladder; treat as PPH if bleeding
2	**Blood loss**	Under 500 mL (calibrated drape / weighed pads)	500 mL or more, or 300 mL or more with an abnormal vital sign → E-MOTIVE bundle

Order	Check	Normal finding to document	Abnormal finding and action
3	**Placenta and membranes**	Complete: all cotyledons, membranes complete, 3 cord vessels, central/eccentric cord insertion	Missing cotyledon or vessels running to a torn membrane edge (**succenturiate lobe**) → exploration/manual removal; 2 vessels → examine the baby
4	**Perineum, vagina, cervix** in good light	Intact, or 1st/2nd-degree tear	3rd/4th-degree, high vaginal or cervical tear → senior, theatre
5	**Rectal examination**	Sphincter intact (pill-rolling), no mucosal tear	Thin or absent sphincter anteriorly → OASI
6	**Vital signs**	Pulse under 100, BP normal, shock index under 0.9	Shock index 0.9 or more → escalate; 1.4 or more → urgent resuscitation and blood
7	**Bladder**	Voided within 6 h	Palpable bladder, unable to void → catheterise
8	**Baby**	Breathing, warm, skin-to-skin, breastfeeding within 1 h	Grunting, cyanosis, cold → neonatal review

Bedside tests

Test	How	What it tells you
Placental examination	Lay it out on a flat surface: maternal side (cotyledons fit together, no ragged area), fetal side and membranes (holes, vessels to the edge), count cord vessels, note cord length and insertion	Completeness → decides on exploration/manual removal
Measured blood loss	Calibrated under-buttock drape, or weigh pads and linen (1 g ≈ 1 mL)	Visual estimates underestimate by up to half; measuring triggers earlier PPH treatment
Shock index	Pulse ÷ systolic BP	0.9 or more abnormal; 1.4 or more urgent intervention
HemoCue	After a large blood loss, or before discharge if anaemic	Baseline for transfusion and iron decisions (a very early Hb underestimates loss)
Bedside clotting test	2 mL venous blood in a plain glass tube	No clot by 7 min or a clot that breaks down → coagulopathy
Rectal examination	Before and after any repair	Sphincter integrity; no suture through the mucosa
Swab, needle and instrument count	Before starting and after finishing a repair	A retained swab causes sepsis and fistula

Laboratory and imaging

A normal delivery needs no laboratory tests. They become relevant when the checks above are abnormal.

Test	When	Expected / abnormal finding	How it changes management
Hb / FBC	PPH, pallor, symptomatic, known anaemia	Hb 11 g/dL or more before delivery; a fall after PPH (may take hours to show)	Iron (ferrous sulphate + folic acid) vs transfusion. HemoCue at HC III/IV; FBC at hospitals

Test	When	Expected / abnormal finding	How it changes management
Group and crossmatch	Blood loss 500 mL or more, shock, haematoma, OASI going to theatre	Compatible units	Transfusion; HC III must refer for blood
Blood group of mother and cord blood	Mother Rh negative	Baby Rh positive	**Anti-D within 72 h**
Coagulation (PT/APTT, fibrinogen, platelets)	Massive PPH, abruption, clotting test abnormal	Fibrinogen under 2 g/L = severe coagulopathy	FFP, cryoprecipitate, platelets. Regional/national referral only; bedside clotting test elsewhere
Wound swab culture	Infected perineal wound	Organism and sensitivity	Tailor antibiotics. Regional/national referral
Pelvic ultrasound	Suspected retained products, haematoma, persistent bleeding with a contracted uterus	Retained tissue; collection	Evacuation or exploration in theatre. HC IV and hospitals, operator-dependent
Endoanal ultrasound / anorectal studies	After OASI with symptoms	Sphincter defect	Plan for future delivery. Rarely available in Uganda (national referral / private)

Red flags

RED FLAG

- **Brisk bleeding** from a tear with a contracted uterus: trauma causing PPH. Apply pressure, call for help, repair promptly in good light.
- **Cannot see the apex** of a vaginal tear or suspect a **cervical/high vaginal tear**: needs a senior, possibly theatre.
- **Sphincter involvement** (3rd/4th degree): senior repair in theatre.
- **Severe perineal or rectal pain** and a swelling after repair, with tachycardia: **haematoma** (can conceal a litre of blood).
- **Urinary retention** after repair: catheterise.
- Later: **fever, foul discharge, wound breakdown** (infection); **faecal or flatus incontinence**; **passage of stool/flatus per vagina** (rectovaginal fistula).
- **Shoulder dystocia** during delivery: HELPERR drill.

RED FLAG

Stop and call for help if: the head delivers and retracts (**turtle sign**: shoulder dystocia), there is a **gush of bright blood with a soft uterus** (PPH), the cord **snaps** during CCT, a **mass appears at the introitus with shock** after CCT (uterine inversion), or the mother collapses.

7. Clinical Acumen

Thinking like a clinician

- **"Bleeding after delivery: is it tone or trauma?"** If the uterus is well contracted and bleeding continues (often bright red, steady), **look for a tear**. If the uterus is soft, rub it up first.

- **"Is this a 2nd or 3rd?"** Only the rectal exam tells you. A perineum that looks like a 2nd-degree tear can hide a torn EAS whose ends have retracted.
- **"Should I cut an episiotomy?"** Only if there is an indication (fetal distress with a tight perineum as the delaying factor, instrumental delivery). Tears usually heal as well as or better than episiotomies.
- **"Suture through the rectum?"** Always finish with a rectal exam.
- **Shock out of proportion to visible bleeding with a firm uterus** → concealed bleeding (haematoma, high vaginal tear).
- **Visual estimates underestimate blood loss**: measure, and trust the vital signs.

Common mistakes (pitfalls)

- Starting below the apex (bleeding continues from retracted vessels above).
- Tying too tightly (pain, necrosis).
- Leaving dead space (haematoma).
- Forgetting the vaginal tampon.
- Failing to count swabs.
- Not giving enough anaesthetic (the woman suffers and moves, and the repair is poor).
- Classifying a tear without a rectal examination.

Worked examples

EXAM FAVOURITE

Worked example 1: the placenta and the bleeding. Twenty minutes after an SVD, a P4 has a firm uterus after oxytocin, but is still trickling. On the placenta, the maternal surface is complete, but **a blood vessel runs across the membranes to a torn hole at the edge**. **Interpretation**: a **succenturiate lobe** has probably been left behind (retained placental tissue). **Action**: call a senior, IV access, second uterotonic, **manual exploration of the uterus** under analgesia with prophylactic antibiotics; watch for PPH.

EXAM FAVOURITE

Worked example 2: the vital signs tell the story. One hour after delivery, a primipara's pad count is "moderate" and the uterus is firm. Pulse has risen from 84 to **118**, BP **98/60** (shock index 1.2), and she has **severe rectal pain**. **Interpretation**: a firm uterus with shock out of proportion to the visible bleeding and rectal pain = **concealed bleeding into a vulval/paravaginal haematoma** (or an unseen high vaginal tear). **Action**: call for help, two IV lines, crossmatch, tranexamic acid 1 g IV (within 3 h of birth), examine in good light; a large or expanding haematoma needs **evacuation in theatre**.

EXAM FAVOURITE

Worked example 3: classifying the tear. After a big baby, you see a perineal tear that reaches the anal margin. On rectal examination, with a finger in the rectum and the thumb on the perineum, the anterior sphincter feels thin, and you can see torn, dark-red external sphincter fibres, but you cannot tell whether more or less than half its thickness is torn. The pale internal sphincter and the rectal mucosa look intact. **Interpretation**: a third-degree tear; because you are unsure between 3a and 3b, **classify it as 3b** (the higher degree, RCOG). **Action**: pressure pad, do not repair yourself, call the obstetrician, repair in theatre under spinal, antibiotics, laxatives, follow-up.

Decision flow when you finish a delivery

1. Uterus contracted? If not, massage and uterotonics first.
2. Placenta complete? If not, explore.

3. Inspect in good light: where is the bleeding from?
4. Rectal exam: is the sphincter intact?
5. **First/second-degree**: repair now (you, under supervision). **Third/fourth-degree or cervical/high vaginal**: call senior; theatre.
6. Count, document, analgesia, counsel.

8. Complications and Troubleshooting

Shoulder dystocia: the HELPERR drill (know it, even as a student)

Letter	Action
H	Call for **Help** (senior, midwife, neonatal help); note the time the head was delivered
E	Evaluate for **Episiotomy** (to make room for internal manoeuvres)
L	**Legs**: McRoberts manoeuvre (hyperflex the mother's thighs onto her abdomen): resolves many cases
P	**Suprapubic Pressure** (not fundal pressure) over the posterior aspect of the anterior shoulder to adduct and rotate it
E	**Enter** the vagina for internal rotation manoeuvres (Rubin II, Woods screw)
R	**Remove the posterior arm**
R	**Roll** the patient onto all fours (Gaskin manoeuvre)

Avoid excessive downward traction on the head (brachial plexus injury, Erb's palsy) and **never fundal pressure** (worsens impaction, risks uterine rupture).

Managing complications

Complication	Management
Bleeding from tear	Pressure, good light, find the apex, suture above it; if cannot see, call senior; treat as PPH (IV access, fluids, tranexamic acid 1 g IV within 3 h of birth)
Vulval/paravaginal haematoma	Small (under 5 cm), stable: analgesia, ice packs, observe. Large or expanding: **evacuate in theatre**, ligate bleeders, drain/pack, catheterise, transfuse if needed, antibiotics
Retained placental tissue (e.g. succenturiate lobe)	Senior, IV access, second uterotonic, manual exploration under analgesia with prophylactic antibiotics
Wound infection	Clean, analgesia, antibiotics (e.g. amoxicillin-clavulanate + metronidazole, or per sensitivity); remove infected sutures if needed
Wound breakdown (dehiscence)	Usually heal by secondary intention after infection clears; resuture only when clean (senior decision)
Retained swab	Foul discharge and fever days later; remove, antibiotics; prevention by counting
Suture through rectum	Remove and resuture; if missed, may cause rectovaginal fistula: refer to fistula surgeon
Urinary retention	Catheterise
Dyspareunia, pain	Assess at 6 weeks; may need revision
Anal incontinence after OASI	Physiotherapy, endoanal ultrasound where available, colorectal/urogynaecology review; discuss mode of next delivery (elective CS if symptomatic or abnormal studies)

Routine postnatal perineal care

1. **Analgesia**: paracetamol **1 g orally every 6-8 hours** (max 4 g/day) plus, if no contraindication (asthma, pre-eclampsia with renal impairment, PPH with coagulopathy), **ibuprofen 400 mg orally every 8 hours** or

diclofenac 50 mg every 8 hours with food, for about 3-5 days.

2. **Hygiene**: wash the perineum with clean water after passing urine or stool, pat dry, change pads every 4-6 hours; salt sitz baths are popular in Uganda (warm water; avoid very strong salt solutions or herbal insertions).
3. **Bladder**: ensure she passes urine within 6 hours; catheterise if not.
4. **Bowels**: fluids, fibre; laxatives after OASI.
5. **Counsel**: danger signs (fever, increasing pain, foul discharge, wound breakdown, incontinence), pelvic floor exercises, and when to resume intercourse (when healed and comfortable, usually after 6 weeks), family planning.

Drug	Dose	Route	Frequency	Duration / notes
Oxytocin (AMTSL)	10 IU	IM	Once, within 1 min of birth	After excluding a second twin
Misoprostol (if no oxytocin)	600 micrograms	Oral	Once	AMTSL alternative
Tranexamic acid (PPH)	1 g over 10 min	IV	Once; second 1 g if bleeding continues after 30 min	Within 3 h of birth
Paracetamol	1 g	Oral	6-8 hourly	3-5 days; max 4 g/day
Ibuprofen	400 mg	Oral	8 hourly with food	3-5 days; avoid in asthma, renal impairment, severe pre-eclampsia
Diclofenac	50 mg (or 100 mg rectal once after repair)	Oral/PR	8 hourly	As above
Lignocaine 1%	Up to about 3 mg/kg (usually 10 mL)	Local infiltration	Once during repair	Aspirate first
Lactulose (after OASI)	10-15 mL	Oral	Twice daily	About 10 days
Antibiotics (after OASI)	e.g. cefuroxime 1.5 g IV + metronidazole 500 mg IV intraoperatively, then oral amoxicillin-clavulanate 625 mg 8 hourly + metronidazole 400 mg 8 hourly	IV then oral	–	About 5-7 days (per local protocol)

Complications by source

- **Of delivery**: perineal/vaginal/cervical tears, PPH, shoulder dystocia, cord avulsion, uterine inversion (cord traction without counter-traction), neonatal injury (brachial plexus, clavicle fracture).
- **Of episiotomy**: extension to OASI, bleeding, haematoma, infection, dehiscence, dyspareunia, poor cosmetic result.
- **Of repair**: pain, infection, dehiscence, retained swab, suture through rectum (rectovaginal fistula), granulation tissue, long-term dyspareunia.
- **Of missed OASI**: faecal/flatus incontinence, perineal body deficiency, rectovaginal fistula, psychosexual problems.

Troubleshooting during the repair

Problem	What to do
Blood from above obscures the view	Vaginal tampon/gauze roll with a tail (count it, remove it)
Cannot see the apex	Better light, assistant to retract; if still not seen, call a senior
Woman in pain during repair	Stop; top up lignocaine within the maximum dose
Suture felt in the rectum on final PR	Remove and redo that layer
Count incorrect	Do not finish; search the vagina, bed and bins before she leaves

9. Documentation and Presenting Findings

Model one-line summary: "A 23-year-old P1 who had an SVD of a live 3.6 kg baby at 03:10 with AMTSL; placenta complete, measured blood loss 350 mL, uterus firm, **second-degree tear with an intact sphincter** repaired with 2-0 polyglactin; vital signs normal."

Problem list: (1) post-SVD, day 0; (2) second-degree perineal tear, repaired; (3) PPH risk period (first 2 hours); (4) breastfeeding and PPFP.

Working diagnosis: "Uncomplicated spontaneous vaginal delivery with a repaired second-degree tear."

Model presentation (postnatal ward round)

"Mrs L.N., a 23-year-old primipara, had a spontaneous vaginal delivery at 03:10 of a live male infant, 3.6 kg, Apgar 8 and 10. AMTSL was given with oxytocin 10 IU IM; the placenta and membranes were complete; estimated blood loss 350 mL. She sustained a **second-degree perineal tear**, confirmed on rectal examination with an **intact anal sphincter**, repaired under 1% lignocaine with **continuous 2-0 Vicryl Rapide** in three layers; rectal examination after repair was normal and swab and needle counts were correct. **Today** she is afebrile, the uterus is well contracted, lochia normal, perineum mildly oedematous, she has passed urine and is breastfeeding. **Plan**: analgesia (paracetamol and ibuprofen), perineal hygiene, pelvic floor exercises, discharge in 24 h with review of the perineum at the postnatal visit."

The delivery and repair note must record: time of birth, sex, weight, Apgar; uterotonic and time; placenta and membranes complete; measured blood loss; degree of tear (with rectal examination findings); anaesthetic and dose; suture material and technique; rectal examination after repair; swab and needle counts; vital signs; plan.

10. OSCE Checklist

Simulated normal delivery

Step	Done well when	Critical?
Introduces self, consent, privacy, companion	–	No
Hand hygiene, PPE, prepares delivery pack, oxytocin and newborn corner (bag and mask checked)	Bag and mask tested before birth	Yes
Cleans and drapes vulva correctly (front to back, one swab per stroke)	–	No
Confirms full dilatation; monitors FHR every 5 min	–	Yes
Supports perineum and keeps head flexed; asks mother to pant at crowning	Head delivered slowly between contractions	Yes

Step	Done well when	Critical?
Wipes face; checks for cord around the neck and manages correctly	No routine suction	No
Waits for restitution and external rotation	–	No
Delivers anterior shoulder (downwards), posterior shoulder (upwards), body by lateral flexion; notes time	Never pulls on the neck	Yes
Dries baby, assesses breathing, skin-to-skin, covers	–	Yes
Palpates for second twin, gives oxytocin 10 IU IM within 1 min	States the twin check aloud	Yes
Delays cord clamping 1-3 min; clamps and cuts correctly	–	No
Performs CCT with counter-traction; delivers placenta and membranes	Only during a contraction	Yes
Checks uterine tone; examines placenta	Counts cord vessels; checks membranes for vessels to the edge	No
Inspects perineum/vagina, does rectal exam, estimates blood loss	Measures rather than guesses	Yes
Disposes of sharps, documents, communicates with mother	–	No

Perineal repair (second-degree)

Step	Done well when	Critical?
Consent, analgesia (lignocaine, aspirate), good light, position	Dose within maximum	Yes
Rectal exam before repair; classifies tear	–	Yes
Identifies apex; first stitch 1 cm above	–	Yes
Continuous non-locking vaginal mucosa to hymenal ring	–	No
Rebuilds perineal body; closes muscle, no dead space	–	No
Subcuticular skin closure	–	No
Rectal exam after repair	–	Yes
Removes tampon; counts swabs and needles	–	Yes
Post-repair advice, analgesia, documentation	–	No

ll. Key Points

High-yield points

- **Controlled delivery of the head** (flex, pant at crowning) is the best tear prevention.
- **Exclude a second twin**, then oxytocin 10 IU IM within 1 minute.
- **Delay cord clamping** 1-3 minutes in a vigorous baby.
- **Episiotomy is restrictive**, mediolateral, about 60 degrees, at crowning.
- **Rectal examination before and after** every repair.
- **Unsure 3a or 3b → call it 3b.** OASI goes to theatre with an experienced surgeon.
- First stitch **1 cm above the apex**; continuous non-locking 2-0 polyglactin.
- **Firm uterus + bleeding = trauma or retained tissue**: look in good light.
- **Measure** blood loss; PPH is 500 mL or more (or 300 mL with abnormal vital signs).
- Shoulder dystocia: **HELPERR**, McRoberts + suprapubic pressure, never fundal pressure.

- **Count** swabs, needles and instruments before and after.

Numbers & doses

Item	Value
AMTSL uterotonic	**Oxytocin 10 IU IM within 1 min** of birth (after excluding a second twin)
No oxytocin	**Misoprostol 600 micrograms orally**
Delayed cord clamping	**1–3 min** in a vigorous baby; ties at about **2 cm and 5 cm** from the umbilicus
FHR in second stage	**Every 5 min** (or after each contraction)
Uterine tone and bleeding checks	**Every 15 min for 2 h**
PPH	**500 mL or more** (or 300 mL or more + abnormal vital sign, WHO/FIGO/ICM 2025) Verify *confirm the 2025 consolidated PPH definition wording*; severe **1,000 mL or more**
Shock index	**0.9 or more** abnormal; **1.4 or more** urgent intervention
Tranexamic acid (PPH)	**1 g IV over 10 min within 3 h of birth**; second 1 g if bleeding continues after 30 min
Retained placenta	Not delivered **30 min** after birth
Episiotomy	Mediolateral, **about 60°** from the midline, at crowning, cut about 3–4 cm
Lignocaine 1%	**10 mg/mL**; max about **3 mg/kg** plain (usually 10 mL is enough)
Second-degree repair	**2-0 polyglactin** (Vicryl Rapide), continuous non-locking; first stitch **1 cm above the apex**
Sphincter repair	**3-0 PDS** (or 2-0 polyglactin), in theatre
3a / 3b / 3c	**Under 50% EAS / over 50% EAS / EAS + IAS**
Post-OASI	Antibiotics, **lactulose 10–15 mL twice daily about 10 days**, catheter 12–24 h, review 6–12 weeks
Placenta	Cord has **3 vessels** (2 arteries, 1 vein)
Void after delivery	Within **6 h**, otherwise catheterise
Haematoma managed conservatively	Under **5 cm** and stable

Exam traps

- **"The perineum looks like a second-degree tear, so it is one."** → Only a **rectal examination** can exclude a torn sphincter whose ends have retracted.
- **"Unsure whether 3a or 3b: call it 3a."** → Wrong. **Classify higher** (3b) so the repair is adequate.
- **"Routine episiotomy for every primigravida prevents tears."** → Wrong. WHO: **restrictive episiotomy** only; it does not prevent OASI.
- **"Bleeding after delivery: give more oxytocin."** → Only if the uterus is soft. **Firm uterus + ongoing bleeding = trauma** (tear, cervix) or retained tissue: look.
- **"Placenta looks complete on the maternal side, so all is well."** → Also check the **membranes for vessels running to a torn edge** (succenturiate lobe).
- **"Pull on the cord to speed up the third stage."** → Only CCT **during a contraction with counter-traction**; traction on a relaxed uterus causes **inversion**.
- **"Shoulder dystocia: apply fundal pressure and pull harder."** → Wrong. **McRoberts + suprapubic pressure**; fundal pressure worsens impaction; hard traction causes Erb's palsy.
- **"Visual estimate of blood loss was 300 mL, so no PPH."** → Visual estimates **underestimate**; measure, and use vital signs (shock index).

Questions seniors ask

Viva questions

1. **Name the instruments in a delivery pack and their uses.** (See the equipment table, section 4.)
2. **Why mediolateral rather than midline?** Lower risk of extension into the anal sphincter.
3. **What angle and when?** About 60 degrees from the midline, at crowning.
4. **Classify a tear with less than 50% of the EAS torn.** 3a.
5. **Why rectal examination after repair?** To exclude a suture through the rectal mucosa (fistula risk) and confirm sphincter integrity.
6. **Why is the first stitch placed above the apex?** To catch retracted bleeding vessels at the apex.
7. **Which suture for the perineum and why?** Rapidly absorbed synthetic polyglactin 2-0: less pain, less need for suture removal than catgut.
8. **Maximum dose of lignocaine without adrenaline?** About 3 mg/kg.
9. **What is the management of a large vulval haematoma?** Evacuation in theatre, haemostasis, catheter, antibiotics, blood as needed.
10. **What are the key steps in shoulder dystocia?** HELPERR: call for help, McRoberts plus suprapubic pressure first; no fundal pressure.

On the labour ward

1. **"What do you check before you give the oxytocin?"** That there is **no second baby** (palpate the abdomen); then oxytocin 10 IU IM within 1 minute.
2. **"Walk me through your checks after the placenta is out."** Uterine tone, blood loss, placenta and membranes (and 3 vessels), perineum/vagina/cervix in good light, rectal exam, vital signs, bladder, baby.
3. **"How do you know the placenta is complete?"** Cotyledons fit together without ragged gaps, membranes are complete (both amnion and chorion), and no vessels run to a torn edge.
4. **"The uterus is firm but she keeps bleeding. What next?"** Trauma until proved otherwise: good light, speculum, find the source and the apex; call a senior for cervical or high vaginal tears; treat as PPH (IV access, TXA).
5. **"Why the first stitch above the apex?"** To catch retracted bleeding vessels at the apex.
6. **"Why a rectal exam after the repair?"** To make sure no suture has gone through the rectal mucosa (rectovaginal fistula) and to recheck the sphincter.
7. **"She has severe perineal pain and a pulse of 120 an hour after repair. What is it?"** A **haematoma** until proved otherwise; examine, resuscitate, and evacuate in theatre if large or expanding.

Mnemonics

- **Tear degrees**: "**1 Skin, 2 Muscle, 3 Sphincter (a-b-c), 4 Mucosa**" ("Some Men Sleep Much").
- **3a/3b/3c**: "**a** = **a** bit of EAS (under half); **b** = **big** bit of EAS (over half); **c** = **complete** plus the internal sphincter."
- **Shoulder dystocia**: **HELPERR** (Help, Episiotomy, Legs, Pressure, Enter, Remove, Roll).
- **Repair order**: "**Apex, Vagina, Muscle, Skin, Rectum, Count**" ("A Very Methodical Surgeon Rechecks Carefully").
- **Suture sizes**: "**More zeros, more slender**" (3-0 is thinner than 2-0).

OSCE tips

- **Talk as you do**: "I am supporting the perineum and keeping the head flexed… I am asking her to pant now."
- Always say "I will palpate for a second twin before giving oxytocin," and "I will do a rectal examination before and after the repair."

- On the suture station: pick up the needle holder correctly (thumb and ring finger), load the needle at the junction of the middle and posterior thirds, and use instrument ties.

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SELF-TEST

Flashcards

26 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	List the core instruments in a standard delivery pack.	**2 artery forceps, cord scissors, episiotomy scissors, needle holder, toothed dissecting forceps, sponge-holding forceps, gallipot, kidney dish**, cord ties/clamps, gauze, drapes.
2	Why do episiotomy scissors have a blunt, angled tip?	To **protect the baby's head** and cut the perineum at the correct angle while two fingers shield the head.
3	RCOG classification of perineal tears?	**1st** skin/vaginal epithelium; **2nd** perineal muscles; **3a** under 50% EAS; **3b** over 50% EAS; **3c** EAS + IAS; **4th** sphincter complex + anorectal mucosa.
4	Unsure whether a tear is 3a or 3b: what do you call it?	**The higher degree (3b)**, to avoid under-treatment.
5	What is a rectal buttonhole tear?	A tear of the **rectal mucosa with an intact anal sphincter**. Not a 4th-degree tear, but must be repaired and followed up (fistula risk).
6	Angle and timing of a mediolateral episiotomy?	About **60 degrees from the midline at crowning** (ends up about 45 degrees after delivery).
7	Indications for episiotomy?	Fetal distress with a tight perineum delaying birth, **forceps** (and some vacuum) deliveries, rigid/scarred perineum, room for shoulder dystocia manoeuvres, selected breech. **Not routine.**
8	Lignocaine 1%: concentration and maximum dose (plain)?	**10 mg/mL**; maximum about **3 mg/kg** without adrenaline (about 18-20 mL for 60 kg). Usually 10 mL is enough. **Aspirate first.**
9	Early signs of lignocaine toxicity?	**Perioral tingling, metallic taste, tinnitus, light-headedness**, then drowsiness, seizures, arrhythmias and cardiac arrest.
10	Preferred suture for a second-degree perineal repair?	**Rapidly absorbable polyglactin 910 (e.g. Vicryl Rapide) 2-0** on a round-bodied needle; continuous non-locking technique.
11	Suture used for anal sphincter repair?	**3-0 PDS** (monofilament polydioxanone, long-lasting) or 2-0 polyglactin; IAS repaired separately end-to-end; EAS end-to-end or overlapping.
12	Where is the first stitch placed in a vaginal tear and why?	About **1 cm above the apex**, to catch retracted bleeding vessels.
13	Why do a rectal exam after perineal repair?	To ensure **no suture has passed through the rectal mucosa** (rectovaginal fistula risk) and to confirm the sphincter.
14	Round-bodied vs cutting needle: which for the perineum?	**Round-bodied (taper)**: it separates rather than cuts tissue (muscle, mucosa). Cutting needles are for tough skin.
15	Post-OASI care package?	**Broad-spectrum antibiotics**, **laxatives** (lactulose), analgesia, catheter 12-24 h, physiotherapy, **follow-up at 6-12 weeks** for continence, counselling for future births.
16	HELPERR for shoulder dystocia?	**H**elp, **E**pisiotomy (evaluate), **L**egs (McRoberts), suprapubic **P**ressure, **E**nter (internal rotation), **R**emove posterior arm, **R**oll to all fours.
17	How are the shoulders delivered in a normal delivery?	After restitution and external rotation: **anterior shoulder by gentle downward traction**, **posterior shoulder by upward traction**, then body by **lateral flexion**. Never pull on the neck.
18	Why ask the mother to pant at crowning?	So the head delivers **slowly between contractions**, allowing the perineum to stretch and reducing tears.

#	Question	Answer
19	Which technique of perineal repair causes least pain?	**Continuous non-locking** sutures for all layers (vs interrupted), with rapidly absorbable synthetic material (Cochrane evidence).
20	A woman has severe perineal pain and tachycardia 2 hours after repair. What must you exclude?	**Vulval/paravaginal haematoma** (can conceal a large blood loss). Examine; large or expanding needs evacuation in theatre.
21	Suture gauge: which is thinner, 2-0 or 3-0?	**3-0 is thinner** (more zeros, more slender).
22	List the "fourth-stage sweep" checks after the placenta is delivered, in order.	**Uterine tone → measured blood loss → placenta and membranes (3 vessels) → perineum, vagina, cervix in good light → rectal exam → vital signs (shock index) → bladder → baby.** Then tone and bleeding every 15 min for 2 h.
23	On the membranes you see a blood vessel running to a torn hole at the edge. Significance?	A **succenturiate lobe** has probably been retained. Call a senior; uterotonic; **manual exploration** under analgesia with antibiotics; watch for PPH.
24	Uterus firm, but bleeding continues after delivery. Most likely cause and action?	**Trauma** (cervical, vaginal or perineal tear) or retained tissue. Examine in good light with a speculum, find the apex, call a senior for cervical/high tears; IV access and **TXA 1 g IV within 3 h** of birth.
25	Before a normal delivery, which findings mean you should call a senior rather than proceed alone?	Malpresentation, head ≥2/5 at full dilatation, moulding +++, abnormal FHR, suspected big baby/shoulder dystocia risk, previous CS or OASI, twins, maternal exhaustion, bleeding or BP ≥160/110.
26	How do you measure (not guess) blood loss at delivery, and what thresholds trigger the PPH bundle?	**Calibrated under-buttock drape**, or weigh pads/linen (1 g ≈ 1 mL). Trigger at **≥500 mL**, or **≥300 mL plus an abnormal vital sign** (WHO/FIGO/ICM 2025); shock index ≥0.9 is abnormal.

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. After a normal delivery, a primipara has a perineal tear through the perineal muscles. On rectal examination about one-third of the thickness of the external anal sphincter is torn; the internal sphincter and rectal mucosa are intact. How should this be classified?

A Third-degree tear, 3a
B Second-degree tear
C Third-degree tear, 3b
D Third-degree tear, 3c
E Fourth-degree tear

2. A junior clerk has just repaired a second-degree tear under supervision at an HC IV. Before the woman leaves the delivery couch, what is the MOST important final step?

A Apply antibiotic ointment to the perineum
B Give ergometrine to reduce bleeding
C Remove all sutures from the skin
D Perform a rectal examination and count swabs, needles and instruments
E Insert a vaginal pack for 24 hours

3. A 19-year-old primigravida is crowning. The FHR has been 90 bpm for 4 minutes, the perineum is tight and is clearly the only thing delaying delivery. What is the MOST appropriate action?

A Wait for spontaneous delivery with the next contraction
B Infiltrate lignocaine and perform a mediolateral episiotomy to expedite delivery
C Apply fundal pressure
D Perform a midline episiotomy without anaesthesia
E Transfer for emergency caesarean section

4. The head of a 4.3 kg baby delivers, then retracts tightly against the perineum. Gentle downward traction does not deliver the anterior shoulder. After calling for help, what is the NEXT step?

A Apply strong traction on the head
B Apply fundal pressure
C Cut the cord around the neck
D Perform a Zavanelli manoeuvre
E McRoberts manoeuvre with suprapubic pressure

5. Two hours after an episiotomy repair, a woman complains of severe, increasing perineal and rectal pressure pain. Pulse is 118, BP 98/60. The uterus is firm and vaginal bleeding is minimal. A tense bluish swelling is seen on the right labium. What is the MOST appropriate management? Advanced

A Oral analgesia and review in the morning
B Oxytocin infusion for uterine atony
C Resuscitate and evacuate the haematoma in theatre with haemostasis
D Remove the episiotomy sutures on the ward
E Apply ice packs only

6. You are about to infiltrate 1% lignocaine for a perineal repair in a 50 kg woman. Which statement is CORRECT?

A Aspirate before injecting; the maximum plain lignocaine dose is about 3 mg/kg (about 15 mL here)
B Up to 50 mL is safe because it is a local injection
C Aspiration is unnecessary in the perineum
D Lignocaine with adrenaline cannot be used in the body anywhere
E The maximum dose is 1 mg/kg

7. Which suture is BEST for repairing the vaginal mucosa and perineal muscles in a second-degree tear?

A Nylon 2-0 on a cutting needle, interrupted
B Silk 0, locking continuous
C Chromic catgut 1, interrupted
D Rapidly absorbable polyglactin 910 (Vicryl Rapide) 2-0 on a round-bodied needle, continuous non-locking
E Polypropylene (Prolene) 3-0

8. After a vacuum delivery at an HC IV, the clerk sees a torn perineum. On rectal exam the external sphincter is completely torn and the pale internal sphincter is also disrupted; the rectal mucosa is intact. There is no experienced obstetrician on site today. What is the BEST course of action?

A The clerk repairs it on the delivery bed with interrupted catgut
B Leave it to heal by secondary intention
C Classify as a second-degree tear and repair
D Wait 6 weeks and then repair
E Control bleeding with a pressure pack, give antibiotics and refer to a hospital for repair in theatre by an experienced surgeon

9. During a delivery, as the head crowns, what is the BEST way to reduce the risk of a perineal tear?

A Ask the mother to push as hard as possible to finish quickly
B Perform a routine episiotomy
C Control the head (keep flexed), support the perineum with a warm compress and ask her to pant so the head delivers slowly
D Apply fundal pressure to avoid a prolonged crowning
E Pull the head out quickly with both hands

10. A woman is bleeding steadily with bright red blood after delivery of a complete placenta. Her uterus is firm and well contracted at the umbilicus. What is the MOST likely cause and the NEXT step?

A Genital tract trauma: inspect the cervix, vagina and perineum in good light and repair
B Uterine atony: give more oxytocin
C Retained placenta: manual removal
D Coagulopathy: give fresh frozen plasma immediately
E Uterine inversion: replace the uterus

11. A student repairing a second-degree tear cannot see the apex of the vaginal laceration because blood keeps obscuring the field, even though the uterus is well contracted. What is the BEST action? Advanced

A Start suturing from the visible part and work downwards
B Pack the vagina and send the woman to the postnatal ward
C Give ergometrine and wait
D Call a senior; improve light, analgesia and exposure (assistant, retractors, a tampon above) to identify the apex before suturing
E Suture the skin only

12. Which is the correct sequence for delivering a baby after the head is born in a normal vertex delivery?

A Immediately pull the head downwards to deliver both shoulders together

B Wipe the face, check for a nuchal cord, await restitution and external rotation, deliver the anterior shoulder downwards, then the posterior shoulder upwards, then the body by lateral flexion

C Rotate the head 180 degrees, then deliver the posterior shoulder first

D Suction mouth and nose deeply, then pull the body out

E Clamp and cut the cord, then deliver the shoulders

13. Three days after a second-degree tear repair, a woman returns with fever of 38.4 degrees C, foul-smelling vaginal discharge and a painful perineum. The wound is intact. The notes do not record a swab count. What should you suspect and do FIRST? Advanced

A Normal lochia; reassure

B Urinary tract infection; give nitrofurantoin

C Malaria only; give artemether-lumefantrine and discharge

D Endometritis only; start IV antibiotics without examination

E Retained swab or vaginal tampon: speculum examination to look for and remove it, then antibiotics

14. Which statement about episiotomy is CORRECT according to WHO 2018 intrapartum recommendations?

A Routine episiotomy is recommended for all primigravidae

B Midline episiotomy is safer than mediolateral in terms of sphincter injury

C Routine or liberal use of episiotomy is not recommended

D Episiotomy prevents all third-degree tears

E Episiotomy should be performed before the head reaches the perineum

15. During a simulated delivery OSCE, immediately after delivering the baby and drying him (he is crying), the candidate reaches for the oxytocin. What must the candidate say or do before injecting it?

A Clamp and cut the cord immediately

B Palpate the abdomen to exclude another fetus

C Deliver the placenta first

D Give vitamin K to the baby

E Perform a vaginal examination

Answers & explanations

> **1. A** · Third-degree tear, 3a
>
> Less than 50% of EAS thickness torn = **3a**. A second-degree tear (B) spares the sphincter. 3b (C) is more than 50% of the EAS. 3c (D) includes the IAS. A fourth-degree tear (E) involves the anorectal mucosa.

> **2. D** · Perform a rectal examination and count swabs, needles and instruments
>
> A **rectal exam** confirms no suture has been placed through the rectal mucosa (fistula risk), and the **count** ensures no retained swab/tampon (sepsis). Antibiotic ointment (A) is not standard. Ergometrine (B) is not indicated for trauma and is contraindicated in hypertension. Absorbable skin sutures are not removed (C). A 24-hour pack (E) is not routine and causes discomfort and retention.

> **3. B** · Infiltrate lignocaine and perform a mediolateral episiotomy to expedite delivery
>
> **Fetal distress with the perineum as the only obstruction** is a clear indication for a **mediolateral episiotomy** (with local anaesthetic). Waiting (A) risks hypoxic injury. Fundal pressure (C) is not recommended (uterine rupture, fetal injury). Midline episiotomy (D) increases OASI risk and cutting without anaesthesia is unacceptable. CS (E) at crowning is slower and more dangerous than an episiotomy.

4. E · McRoberts manoeuvre with suprapubic pressure

This is **shoulder dystocia**. After calling for help, **McRoberts plus suprapubic pressure** resolves most cases. Strong traction (A) causes brachial plexus injury. **Fundal pressure** (B) worsens impaction and can rupture the uterus. Cutting a nuchal cord (C) removes the baby's only oxygen supply before delivery. Zavanelli (D) is a last-resort manoeuvre.

5. C · Resuscitate and evacuate the haematoma in theatre with haemostasis

Severe pain, a tense swelling and **haemodynamic compromise** with a contracted uterus = **large vulval/paravaginal haematoma**, which can hide a litre or more of blood. Resuscitate (IV access, crossmatch), then **evacuate in theatre**, ligate bleeders, catheterise. Analgesia alone (A) or ice alone (E) are only for small, stable haematomas. Oxytocin (B) is for atony, but the uterus is firm. Removing sutures on the ward (D) without anaesthesia and light is unsafe and may not reach the bleeding point.

6. A · Aspirate before injecting; the maximum plain lignocaine dose is about 3 mg/kg (about 15 mL here)

Lignocaine 1% = 10 mg/mL; the plain maximum is about **3 mg/kg** = 150 mg = **15 mL** for 50 kg. **Aspirate** to avoid intravascular injection. 50 mL (B) would be 500 mg, toxic. The perineum is vascular, so aspiration matters (C). Adrenaline-containing lignocaine is widely used elsewhere (D) (max about 7 mg/kg). 1 mg/kg (E) is too low (that is closer to the IV antiarrhythmic dose).

7. D · Rapidly absorbable polyglactin 910 (Vicryl Rapide) 2-0 on a round-bodied needle, continuous non-locking

Evidence favours **rapidly absorbable synthetic** sutures with a **continuous non-locking** technique (less pain, fewer suture removals). Nylon (A), silk (B) and Prolene (E) are non-absorbable and would need removal and cause discomfort. Chromic catgut (C) causes more pain and tissue reaction; size 1 is too thick. Locking stitches (B) strangulate tissue.

8. E · Control bleeding with a pressure pack, give antibiotics and refer to a hospital for repair in theatre by an experienced surgeon

This is a **3c tear** (OASI). It needs **repair in theatre** with anaesthesia, good light and an experienced surgeon; primary repair gives the best chance of continence. A student repairing it on the delivery bed (A) risks a failed repair. Leaving it (B) or delaying (D) causes incontinence and fistula. Calling it second-degree (C) is misclassification and leads to inadequate repair.

9. C · Control the head (keep flexed), support the perineum with a warm compress and ask her to pant so the head delivers slowly

Slow, controlled delivery of the head with flexion, perineal support/warm compress and panting reduces tears (including OASI). Forceful pushing (A), fundal pressure (D) and pulling (E) cause rapid, uncontrolled expulsion and tears. Routine episiotomy (B) does not prevent severe tears and is not recommended.

10. A · Genital tract trauma: inspect the cervix, vagina and perineum in good light and repair

Bleeding with a **firm, contracted uterus** and a complete placenta points to **trauma** (cervical, vaginal or perineal tears): inspect and repair. Atony (B) gives a soft uterus. The placenta is complete (C). Coagulopathy (D) is possible but less likely first; check a bedside clotting test while you look for tears. Inversion (E) presents with an absent fundus and shock.

11. D · Call a senior; improve light, analgesia and exposure (assistant, retractors, a tampon above) to identify the apex before suturing

You must **find the apex** and place the first stitch above it to control retracted bleeding vessels. With poor exposure, **call for help**, improve light and analgesia, use retractors and a tampon with a tail. Starting below the apex (A) leaves a bleeding point that forms a haematoma. Sending her away packed (B) delays repair. Ergometrine (C) treats atony, not trauma. Skin-only closure (E) hides a bleeding, unrepaired tear.

12. B · Wipe the face, check for a nuchal cord, await restitution and external rotation, deliver the anterior shoulder downwards, then the posterior shoulder upwards, then the body by lateral flexion

This is the standard sequence respecting the **mechanism of labour**. Pulling immediately (A) before external rotation risks shoulder impaction and brachial plexus injury. Forced rotation (C) is wrong except in specific manoeuvres. Deep routine suction (D) is not recommended (bradycardia, laryngospasm). Clamping the cord before the body is born (E) is only for a tight nuchal cord that prevents delivery.

13. E · Retained swab or vaginal tampon: speculum examination to look for and remove it, then antibiotics

Foul discharge and fever after a repair with **no documented count** strongly suggest a **retained swab/tampon**; examine with a speculum and remove it, then treat infection. Reassurance (A) is dangerous. UTI (B) and malaria (C) do not explain foul discharge (though test for malaria in febrile patients). Endometritis (D) is possible, but you must examine to find a removable cause.

14. C · Routine or liberal use of episiotomy is not recommended

WHO recommends **against routine or liberal episiotomy**. It should be used only for clear indications. Routine use in primigravidae (A) increases trauma. Midline episiotomy (B) has a **higher** risk of OASI. It does not prevent all severe tears (D). Episiotomy is cut at **crowning** (E), not earlier (more bleeding, no benefit).

15. B · Palpate the abdomen to exclude another fetus

The key safety step before a uterotonic is **excluding an undiagnosed second twin** by abdominal palpation. Immediate cord clamping (A) is not recommended for a crying baby (delay 1-3 min). The placenta (C) is delivered after the uterotonic by CCT. Vitamin K (D) is given later in the first hours. A VE (E) is unnecessary here.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: Simulated delivery OSCE station

OSCE station at UCU skills lab: 'Mrs K, a 24-year-old primigravida at 40 weeks, is fully dilated with the head visible at the introitus. Conduct the delivery on the mannequin, including the third stage, talking the examiner through your actions.'

1. What do you do before the baby is born?

> Introduce yourself, consent, privacy and companion; **hand hygiene, apron, goggles, sterile gloves**; check the delivery pack; draw up **oxytocin 10 IU**; prepare the newborn corner (2 warm towels, cap, **bag and mask size 1 checked**, suction bulb, clock); position; clean and drape the vulva front to back.

2. Describe how you deliver the head and shoulders.

> Support the perineum with a pad/warm compress; keep the head **flexed** with gentle pressure on the occiput; at crowning ask her to **pant**; let the head extend slowly. **Wipe the face**, check for **nuchal cord**, wait for **restitution and external rotation**; deliver the **anterior shoulder downwards**, **posterior shoulder upwards**, then the body by **lateral flexion**; note time of birth.

3. The baby is born crying. What do you do in the first minute?

> Place on the mother's abdomen, **dry thoroughly** (head, body), remove the wet towel, **skin-to-skin**, cover and cap. **Palpate the abdomen to exclude a second twin**, then give **oxytocin 10 IU IM** into the thigh within 1 minute. Delay cord clamping 1-3 min, then clamp at about 2 and 5 cm and cut.

4. Describe the delivery of the placenta and what you check.

> Wait for a contraction; **counter-traction** above the pubis with one hand, **steady downward traction** on the cord with the other; stop and retry if it does not descend in 30-40 s. Deliver the placenta with both hands, twist the membranes. **Massage/check the fundus**. Examine the **placenta** (all cotyledons, membranes complete, no vessels to torn edge, 3 cord vessels).

5. What do you do after the placenta is out?

> Check uterine tone every 15 min for 2 h; **inspect the perineum, vagina (and cervix if bleeding)** in good light; **rectal exam**; estimate blood loss; repair tears; clean and make comfortable; help breastfeeding; dispose of sharps; decontaminate instruments; document time of birth, sex, Apgar, oxytocin, placenta, EBL, perineum.

Take-away: Narrate each step, keep the head flexed and controlled, exclude a second twin before oxytocin, and finish with a perineal and rectal examination.

Case 2: Perineal tear at Mengo after a big baby

Mrs E.N., 27, P1, delivered a 4.1 kg baby after a 2.5 h second stage at Mengo Hospital. The placenta is complete and the uterus is firm. Estimated blood loss is 400 mL and there is a perineal tear.

1. How do you examine the tear systematically?

> Explain and consent; good light, lithotomy, analgesia; inspect labia, periurethral area, vaginal walls and the apex; perineum; cervix with a speculum if bleeding continues; then **rectal examination** with pill-rolling of the sphincter between finger and thumb to assess the EAS/IAS and look for mucosal tears.

2. The perineal muscles are torn; the EAS is intact on pill-rolling; the rectal mucosa is intact. Classify and plan.

> **Second-degree tear.** Repair on the delivery bed under **1% lignocaine** (aspirate, about 10 mL), with an assistant and good light.

3. Describe the repair.

> Tampon with tail high in the vagina if needed. **First stitch 1 cm above the apex**; **continuous non-locking 2-0 Vicryl Rapide** for vaginal mucosa to the hymenal ring; bring the needle out at the fourchette; close the **perineal muscles** (rebuild the perineal body, no dead space); **subcuticular skin**; knot buried in the vagina. **Remove the tampon, rectal exam, count swabs and needles.**

4. What postnatal care and advice will you give?

> **Paracetamol 1 g 6-8 hourly and ibuprofen 400 mg 8 hourly** with food (if no contraindication) for 3-5 days; perineal hygiene, change pads, pat dry; pass urine within 6 h; fluids and fibre; pelvic floor exercises; danger signs (fever, increasing pain, foul discharge, wound breakdown, incontinence); review at the postnatal visit; family planning counselling.

5. If instead the EAS had been torn more than 50%, what would change?

> **3b tear (OASI)**: call the obstetrician; repair in **theatre** under spinal/GA with **3-0 PDS** (EAS overlapping or end-to-end); **broad-spectrum antibiotics**, **lactulose**, analgesia, catheter 12-24 h, physiotherapy, and **follow-up at 6-12 weeks** for continence; counsel regarding future delivery.

Take-away: Every tear is classified only after a rectal exam. Second-degree tears are repaired in layers with continuous non-locking synthetic sutures; OASI goes to theatre with an experienced surgeon.

Case 3: Post-repair collapse at an HC IV

At an HC IV, Ms H.A., 20, P1, had an episiotomy repaired by an intern 90 minutes ago. She now complains of severe perineal pain and an urge to open her bowels. Pulse 124, BP 94/58, pale. The uterus is firm at the umbilicus and pad loss is small.

1. What are your differentials?

> **Vulval/paravaginal (or ischiorectal) haematoma** (most likely: pain, rectal pressure, shock with minimal external bleeding); concealed bleeding from an unrepaired vaginal/cervical tear; less likely uterine atony (uterus firm), uterine rupture, or broad ligament haematoma.

2. What will you do immediately?

> **Call for help.** ABC, oxygen, **two large-bore IV cannulas**, bloods (Hb, grouping and crossmatch 2 units, bedside clotting test), **Ringer's lactate/normal saline 1 L** rapidly, catheterise, monitor vitals every 15 min. **Tranexamic acid 1 g IV** if within 3 h of birth. Examine the perineum and vagina (and rectum) gently.

3. You find a tense 8 cm bluish swelling of the right lateral vaginal wall extending towards the ischiorectal fossa. What is the definitive management?

> **Evacuation in theatre** under spinal/GA: incise over the haematoma (via the vagina), evacuate clots, **ligate bleeding vessels**, close dead space (or drain), vaginal pack for 12-24 h, catheter, **antibiotics**, transfuse as needed. If bleeding continues and the source is high, senior/surgical help.

4. What could have prevented this?

> Placing the **first stitch above the apex**, obliterating **dead space**, good haemostasis, adequate light and help; and **observing the woman every 15 min for 2 h** (pulse, BP, pain) so the haematoma was detected before shock.

5. How would you present this at the morning meeting?

> 'Ms H.A., 20, P1, day 1 after SVD with a mediolateral episiotomy, developed severe perineal pain and hypovolaemic shock (P 124, BP 94/58) 90 minutes after repair with a firm uterus. A 8 cm right paravaginal haematoma was evacuated in theatre, bleeders ligated, 2 units transfused. Now stable, catheterised, on antibiotics and analgesia. Plan: monitor Hb, remove pack and catheter at 24 h, iron supplementation, review of repair technique by the team.'

Take-away: Shock with a firm uterus and little visible bleeding after perineal repair is a haematoma until proven otherwise; resuscitate and evacuate in theatre.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.